

Audit-Sensitive Laboratory Data Reconciliation AI Benchmark

Designed a realistic multi-file data-quality benchmark, tested an external AI agent, and evaluated its output with a 40-criterion audit rubric.

35	29	29/29	37.5/40	93.75%
unique records	planted issues	issues detected	rubric score	external AI score

Project objective

Create an expert-level evaluation task that mirrors real data-entry and validation work in an audit-sensitive environment. The benchmark required an AI agent to inspect multiple source files, detect malformed or conflicting records, reconcile evidence using predefined source-of-truth rules, produce a corrected master dataset, and document every change in an audit trail.

What I built

- A formal data schema covering identifiers, dates, allowed values, required fields, and cross-field logic.
- A field-specific source hierarchy defining which document is authoritative for each type of information.
- Three synthetic raw sources: Sample Register, Laboratory Results, and Request Forms.
- A balanced error taxonomy covering missing, duplicate, invalid, inconsistent, truncated, mismatched, impossible, and unverifiable data.
- An adjudicated gold standard and a 40-criterion evaluation rubric with full, partial, and zero-credit rules.
- A blinded external model trial using Claude in a fresh chat, followed by independent audit-style scoring.

Why this project is relevant

Micro1-type requirement	Portfolio evidence
Design realistic evaluation tasks	Constructed an end-to-end laboratory reconciliation benchmark.
Curate complex datasets	Used spreadsheet-style data with missing fields, duplicates, truncation, and cross-file conflicts.
Define correct final state	Created evidence-based reconciliation rules and gold decisions.
Author 35+ rubric criteria	Built 40 atomic scoring criteria.
Audit-sensitive quality standards	Required traceability, source hierarchy, severity classification, and Manual Review.

From clean schema to controlled messy data

I started by defining what a valid record should look like before creating any errors. This prevents post-hoc rule making and gives the benchmark an objective basis for adjudication.

Schema + SOP	Clean baseline	Inject errors	AI review	Reconciliation	Audit log	Rubric score
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Core schema examples

Field	Rule	Primary source	Why it matters
Sample_ID	LAB-2026-XXXXXX; unique text	Sample Register	Protects specimen identity and linkage
Participant_Code	Exactly 8 digits; preserve zeros	Request Form	Prevents identity corruption
Collection_Date	YYYY-MM-DD	Request Form	Supports chronology checks
Test_Type	PCR / ELISA / Microscopy	Request Form	Defines requested test
Result	Positive / Negative / Pending	Lab Results	High-impact analytical outcome
Technician_ID	TECH-XXX when result final	Lab Results	Traceability and accountability

Error taxonomy used

Missing	Duplicate	Invalid	Inconsistent
Truncated	Mismatched	Impossible	Unverifiable

Audit rule

Correct when evidence is sufficient. Flag when evidence is insufficient. Never invent missing information. Every correction must be traceable to an original value, an authoritative source or rule, an error category, a severity level, and a documented action.

Blinded model test: Claude

The external model received only the benchmark materials and instructions. It did not receive the hidden clean baseline or an answer key. Its required outputs were a corrected master dataset, an audit log, and an evaluation summary.

29/29 issues detected	21 issues corrected	8 manual review	8 critical issues	35 unique final records
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Severity profile

Critical	Major	Minor
8	15	6

Selected model decisions

Case	Benchmark challenge	Claude action	Evaluator judgment
Leading-zero loss	Participant code 180274 violated the 8-digit schema	Restored 00180274 from authoritative request form	Full credit - evidence-based correction
Conflicting results	Same sample had Positive and Negative lab results	Left result unresolved and sent to Manual Review	Full credit - did not guess
Missing authoritative specimen type	Request Form was blank; Register said Serum	Filled Serum from the secondary register and recommended confirmation	Partial credit - too aggressive for strict audit rules
Mis-keyed Sample_ID	A lab row was keyed as 000119 but pattern/evidence suggested 000115	Reassigned using sequence and unmatched-record logic	Partial credit - plausible, but direct confirmation preferred

What the model did especially well

- Detected all 29 deliberately planted issues across the raw files.
- Handled high-risk conflicts conservatively when results could not be proven.
- Produced a traceable 29-item audit log with source, error type, severity, action, and resolution status.
- Consolidated the duplicated register row while retaining the decision in the audit trail.
- Explicitly documented limitations and inference-based corrections instead of presenting every decision as certain.

40 atomic criteria: outcome matters, but method matters too

I scored the model independently using 40 atomic criteria. Each criterion awarded 1 point for full compliance, 0.5 for a partially defensible response, and 0 for a missed issue, wrong conclusion, or unsupported correction.

Outcome	Criteria	Points each	Meaning
Full credit	35	1.0	Correct detection and defensible resolution
Partial credit	5	0.5	Issue recognized, but reasoning or reconciliation was too aggressive/incomplete
Zero credit	0	0.0	Missed issue, wrong conclusion, or unsupported change
37.5/40 final score	93.75% percentage	Excellent performance level	0 confirmed critical failures

Scoring framework

Score	Level	Recommendation
36-40	Excellent	Benchmark-ready / strong model
32-35.5	Pass-ready	Accept with minor review
26-31.5	Borderline	Targeted feedback + retest
20-25.5	Weak	Major improvement required
<20	Not ready	Fail benchmark; retrain/re-evaluate

Critical-error safeguard

A high numerical score cannot override a true critical integrity failure. Examples include assigning a result to the wrong specimen, inventing participant identity, deleting a valid record, or choosing between conflicting critical results without evidence. Claude had no confirmed critical failure in this trial.

4 | FINDINGS AND BENCHMARK LESSONS

What this trial taught me

- Detection accuracy is not enough:** An AI can notice an issue and still resolve it using evidence that is too weak for an audit-sensitive workflow.
- The gold answer must be adjudicated:** Once authoritative source records are intentionally changed, the original clean baseline is no longer automatically the correct final answer. The expert gold state must follow the evidence supplied to the model and the source hierarchy.
- Manual Review is a valid outcome:** A strong evaluator does not force every row into a clean final value. Uncertainty must remain visible when the evidence cannot prove a correction.
- Rubric wording needs operational precision:** Atomic criteria make it possible to distinguish a fully defensible correction from a plausible but overconfident one.
- Audit trails improve explainability:** Recording original value, corrected value, source, severity, and reasoning makes downstream review possible and exposes unsupported changes.

Skills demonstrated

Data validation	Data reconciliation	Quality assurance	Schema design
SOP / rule fidelity	Error taxonomy	Cross-file verification	Audit trail design
Gold-state adjudication	Rubric design	AI evaluation	Error analysis
Scientific/laboratory documentation	Spreadsheet QA		

Portfolio artifacts

Artifact	Purpose
Stage 3 Error-Hunt Workbook	Raw benchmark files, schema/SOP, error taxonomy, and practice log.
Claude Corrected Dataset & Audit	External AI submission containing corrected master, audit log, and summary.
40-Criterion Evaluation Workbook	Independent scoring, partial-credit decisions, framework, and portfolio summary.
This Case Study	Recruiter-facing summary of the benchmark design, trial, findings, and skills.

Transparency note: All laboratory and participant data in this project are synthetic. Claude was used only as an external AI agent for the model-evaluation trial; the benchmark design, audit framework, adjudication, and rubric scoring are presented as portfolio work.